D1. Patient journey & success snapshot

A regional provider network needs better referral coordination across an EHR, portals, scheduling, diagnostic and payer systems, partners, and manual work—without disrupting care or replacing the EHR in Phase 1.

Problem and patient journey

Referral information, ownership, and status are fragmented across systems and hand-offs. **15–20% of referrals require manual follow-up or rework**, contributing to patient delays, duplicate calls, unclear next steps, and difficult-to-detect referral leakage.

1. Primary-care team initiates a referral and supplies clinical context.
2. Referral crosses EHR, portal, scheduling, payer, and partner workflows; information may be incomplete or unavailable.
3. Care coordinators chase missing information, status, and scheduling/eligibility hand-offs.
4. Specialist/partner acts; status and completion are not consistently visible.
5. Patient may face delay, repeated calls, or unclear next steps.

Constraints

Retain EHR in Phase 1 · Initial usable capability expected in ~12 weeks · Normal patient care continues · Minimize, secure, and audit PHI · Govern and review AI outputs · Support standards and future partners · Patient-facing automation must offer safe human escalation.

Assumptions to validate

1. One clinically approved referral pathway can be bounded for the first release.
2. EHR and priority partners provide supported interfaces for minimum referral/status exchange.
3. Referral identity/status can be shared without moving clinical-record authority.
4. Operational records can establish rework and turnaround baselines.
5. Clinical/privacy owners can define human escalation and permitted AI use before runtime AI.

Safety and privacy

- Clinical, urgency, eligibility, and consequential care decisions remain with authorized people.
- Minimize PHI in integrations, logs, dashboards, tests, and any AI flow; prototype with synthetic data.
- Never send identifiable health information to an unapproved AI service.
- Provide ownership and escalation for missing, delayed, conflicting, or urgent referrals.
- A prototype is not clinical validation or evidence of regulatory compliance.

Success measures

Targets are not specified. Establish baselines and agree targets with accountable clinical/operations owners.

Measure	Definition	Baseline	Target	Source / owner
Manual follow-up / rework	Referrals needing manual follow-up or rework ÷ agreed cohort.	15–20%; validate cohort/period.	TBD with owners.	Referral/work-queue records; TBD.
Referral turnaround	Agreed referral start to completion/scheduling event; cohort and statistic TBD.	Not provided.	TBD.	EHR/referral/scheduling timestamps; TBD.
Status / ownership visibility	Sampled in-scope referrals with retrievable status and owner across pathway.	Not provided.	TBD.	EHR/orchestration/partner sample; TBD.
Referral leakage / incomplete hand-off	Items not acknowledged, scheduled, completed, or dispositioned in agreed window.	Not provided.	TBD.	Referral/scheduling/partner records; TBD.
Duplicate patient contact	Sampled referrals with repeated avoidable calls for the same status/next step.	Not provided.	TBD.	Contact/coordinator records; TBD.
Hand-off audit completeness	Sampled hand-offs with status, timestamp, actor/system, and required outcome.	Not provided.	TBD.	Audit records; privacy/audit owner TBD.

Open: first pathway/partner scope, actual interface profiles, clinical exception paths, status semantics, privacy/consent/retention, and baseline definitions.

D2. Architecture options & decision

Keep the EHR, preserve care continuity, protect PHI, improve referral visibility and partner interoperability, and start with a pathway that can be safely bounded. Standards/profile readiness and actual system interfaces remain unknown.

Option A · Referral orchestration hub

Add orchestration around the EHR for an agreed referral pathway. Track coordination status/ownership, route exceptions to coordinators, and connect EHR/partners through replaceable adapters using standards-based interfaces where each endpoint supports them.

Strengths: Better cross-network status and hand-off visibility; isolates partner-specific integration changes.

Trade-offs: New PHI-processing boundary and operations; needs shared identity/status semantics, clinical ownership, partner readiness, and reconciliation. Avoid duplicate/conflicting worklists.

Option B · EHR-centered workflow enhancement

Improve referral workflow/status primarily within the existing EHR and use targeted direct interfaces to priority partners. The EHR remains the main coordination surface.

Strengths: Keeps more work near the clinical record and may limit new workflow surfaces.

Trade-offs: EHR is expensive/risky to modify; partner integrations may remain coupled and cross-network status may be incomplete. Depends on EHR configuration and vendor constraints.

Trade-off comparison

Criterion	Option A · orchestration	Option B · EHR-centered
Phase 1 / usability	One pathway and thin integration slice; depends on endpoint support and agreed semantics.	Bounded EHR workflow; depends on safe configuration and partner interfaces.
Care continuity	EHR retained; coexistence, fallback, and careful routing required.	EHR remains central; change control and rollback required.
Cross-network visibility	Stronger fit if partners exchange reliable IDs, statuses, and outcomes.	Strongest within EHR; partner visibility may remain fragmented.
Interoperability	Shared boundary can reduce point-to-point growth; standards support is endpoint-specific.	Direct selected integrations may be quick but can preserve coupling.
Privacy / audit	New PHI boundary needs explicit minimization, access, audit, and retention.	Uses established EHR controls, but interfaces and additional workflow still need governance.
Operating model	Explicit coordinator queue; avoid duplicate/conflicting EHR worklists.	Workflow close to clinical record; external hand-offs may remain manual.
Partner growth	Adapters support incremental onboarding; throughput/profile fit to validate.	May need more EHR changes or partner-specific interfaces.

Conditional preference

> **Prefer Option A** for one bounded, clinically approved pathway, subject to viable supported interfaces, agreed status/source ownership, partner readiness, privacy review, and safe coexistence.

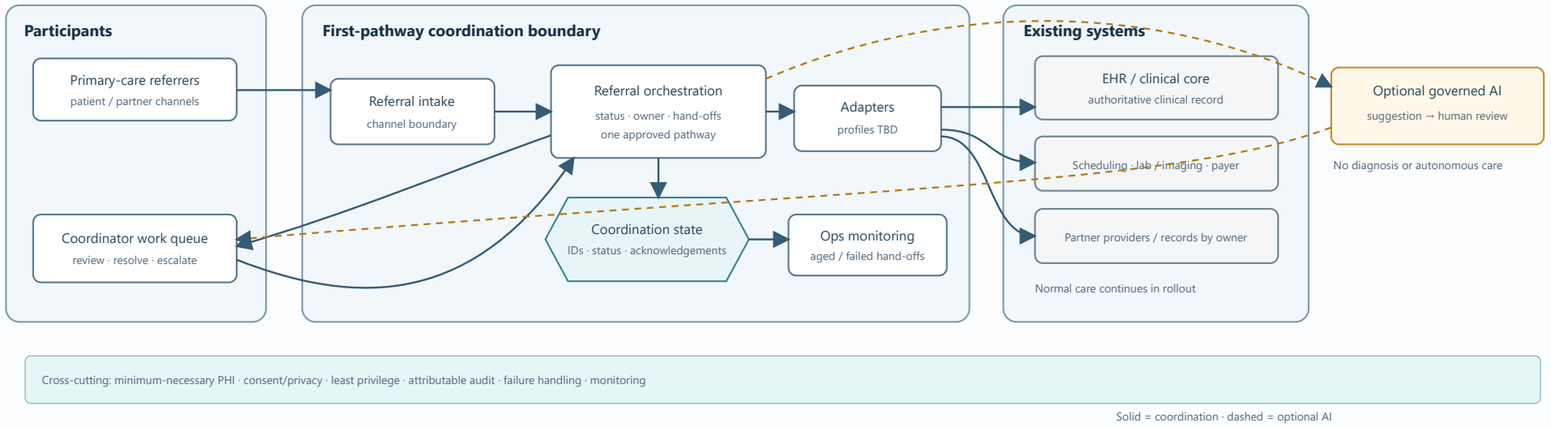
It addresses network-level visibility, hand-offs, and partner integration changeability more directly. Discovery compares supported FHIR workflow profiles, existing HL7 messages and approved proprietary adapters: shared semantics versus endpoint readiness and mapping effort. No standard/profile is yet committed.

Revisit if

- Required interfaces are unavailable or unsafe: narrow scope or start with supported targeted integration.
- No clinical/operational workflow owner or status agreement exists: defer orchestration.
- A small EHR-centered change is demonstrably safer and less disruptive for the first pathway.
- Partner readiness, privacy approval, or safe escalation cannot be established.
- Meeting ~12 weeks would threaten care continuity: reduce scope or resequence.

D3. Target-state architecture

One-pathway referral orchestration around the retained EHR, with governed partner integration, coordinator exception handling, minimum-necessary data, and safe human escalation.



Components and data boundary

- Orchestrator owns referral coordination state, not clinical records.
- EHR remains authoritative for clinical documentation; partner/scheduling/diagnostic systems retain their record ownership.
- Adapters map approved interactions; exact endpoint profiles and standards support are to validate.
- Monitoring tracks delayed/failed hand-offs using minimum necessary data.

Proposed exchange and ownership

Source organization + referral ID identify the referral; status events carry unique event ID, source, version and event time. EHR owns clinical content; hub owns coordination state. Receipt never means clinical acceptance. Missing data/timeout go to coordinator; retry needs actor/reason. Late receipt reconciles; conflicts require review. Validate patient matching, event ordering and partner replay contracts.

Safety, privacy, and resilience

- Agree PHI purpose, minimization, consent, access, audit, retention, and downtime controls before pilot.
- Define matching, duplicates, timeouts, retry, reconciliation, partner acknowledgement, and safe fallback.
- Clinical urgency, ambiguous referrals, and patient-facing exceptions go to authorized humans.
- AI is optional; no diagnosis or autonomous clinical action; approved inputs and human review required.

D4. Focused POC / validation

A standalone synthetic-data prototype tests administrative referral completeness and coordinator follow-up—not clinical triage or FHIR conformance.

Question and hypothesis

> **Question:** Can a thin coordination boundary retain ownership through missing fields, duplicate intake, receipt failure and human-approved retry without making a clinical decision?

Hypothesis: Explicit states, receipt semantics and attributable retry decisions prevent duplicate tasks and silent loss in the synthetic workflow.

Scope and safeguards

- Five fictional browser records; required-field checks plus duplicate intake, receipt, timeout, human-approved retry and decision history.
- Missing data creates administrative coordinator follow-up only.
- No PHI, EHR/partner connection, network request, AI, diagnosis, urgency, eligibility, or clinical appropriateness logic.
- Payload is a proposed coordination contract, not a FHIR resource or validated profile.

Run the prototype

File: `poc-referral-validation.html` in this healthcare output folder.

Keep `referral-workflow.js` beside the HTML. Open in a browser, load intake cases, send a complete referral and simulate receipt. Send the silent-partner case, advance three ticks, record reviewer/reason, approve retry and acknowledge. Replay the first referral to inspect duplicate handling.

Executed validation and learning

8 workflow engine tests passed; assertion outcomes and execution timestamp are in validation-results.json. Browser smoke checks also passed intake, receipt, timeout, human retry and duplicate handling; see browser-validation.json.

Case	Expected / actual
Complete intake/send/receipt	Receipt state and coordinator owner / PASS
Missing ID, specialty or receiver	Owned follow-up; send blocked / PASS
Duplicate intake	One record, first payload retained / PASS
Timeout	Owned escalation, no completion / PASS
Human retry	Actor/reason required; same identity / PASS
Early/duplicate receipt	Rejected, no extra receipt event / PASS
Late receipt	Overdue item reconciled / PASS
Provenance	Ordered actor/reason history / PASS

Learning: explicit receipt semantics, duplicate identity and timeout ownership are testable. Acknowledgement is distinct from clinical acceptance/completion. Three mock ticks are not a production SLA.

What remains unproven

- FHIR release/profile conformance, terminology, patient matching, and real EHR/partner integration.
- Consent, privacy/security controls, authorization, durable audit, retries, reconciliation, scale, or downtime behavior.
- Real workflow usefulness, clinical safety, clinical outcome improvement, or AI performance.

Decision effect: supports usability of an administrative hand-off concept only; D2 remains conditional on pathway, interface, and clinical/privacy validation.

D5. Delivery, safety & risk plan

Sequence the conditional orchestration direction through clinical, interface, privacy, and continuity gates. The ~12-week expectation is a planning constraint, not a guarantee.

Now / Next / Later

Phase	Outcome and work	Exit gate / dependency	Owner
Now · weeks 1–4 proposed	Select pathway/partners; map journey, exceptions, status, and owners; inventory interfaces/data; establish baseline; agree PHI, consent, audit, safety, downtime, and patient communication controls.	Clinical/ops approve bounded workflow; interface owners confirm exchange; privacy/security approve handling. Otherwise narrow/revisit.	Clinical pathway, referral ops, EHR/partner, privacy/security; TBD.
Next · weeks 5–12 proposed	Thin orchestration slice, coordinator queue, replaceable integration, status/acknowledgement history, monitoring. Test matching, duplicates, retries, access/audit, downtime, fallback, rollback; pilot with continuity.	Agreed outcomes; safety/privacy/interface review; safe fallback demonstrated; approval to expand. Weeks 5–8 build; 9–10 safety/partner/fallback tests; 11–12 bounded pilot. Discovery failure triggers scope reduction/replan.	Delivery, clinical owner, integration, operations, privacy/security; TBD.
Later · evidence-led expansion	Review measures/incidents; add specialties/partners incrementally; improve mappings/reconciliation. Consider AI only after approved data use, evidence, human capacity, governance, and monitoring.	Current pathway stable, controls effective, partner operations ready, outcomes understood.	Clinical/product, platform, operations, governance; TBD.

Top five risks and mitigations

Risk / consequence	Mitigation and early trigger	Owner
Clinical safety / unclear escalation: delay or unsafe referral hand-off.	Clinical owners define urgency, ownership, exception response, downtime; test ambiguous/failure cases. **Trigger:** an in-scope referral can be ownerless or lack safe escalation.	Clinical safety owner; TBD.
Interfaces/profile unsupported: status cannot be exchanged; scope/timing at risk.	Discover contracts/profile support early; test thin non-production flow with synthetic data. **Trigger:** IDs/status cannot be exchanged or partner cannot acknowledge.	EHR/partner integration; TBD.
Identity/status/source disagreement: duplicates or conflicting referral state.	Agree matching/source ownership/status mappings/reconciliation; uncertain matches get human review. **Trigger:** source disagreement or ambiguous identity.	Clinical informatics/data owners; TBD.
PHI/privacy/audit controls inadequate: disclosure or incomplete accountability.	Review purpose, minimization, access, retention, audit before pilot; restrict logs. **Trigger:** unapproved data flow, unclear consent/purpose, missing audit evidence.	Privacy/security/compliance; TBD.
Partner adoption/continuity failure: unmonitored referrals or stale parallel work.	Co-design, define acknowledgement/fallback, train users, monitor aged items. **Trigger:** partner readiness unknown or referrals outside monitored workflow without owner.	Referral operations/partner owner; TBD.

AI-use reflection

In this revision Codex generated workflow code, executable assertions, a proposed exchange contract and relative roadmap. Review identified shallow failure validation, unsupported approval wording and generic AI reflection; these were corrected. Automated tests verify deterministic behavior. Clinical, partner and privacy approval remain pending; no such human sign-off is recorded. Earlier gateway usage/cost evidence is unavailable. This POC used deterministic checks, not an AI model; it provides no real interface, clinical, privacy or model-performance evidence. Clinical decisions, PHI approvals, and risk acceptance remain human-owned.

Guardrails

- Keep EHR and patient-care processing in place.
- Bound the first pathway and partner scope.
- Separate clinical records from coordination state.
- Never equate completeness, acknowledgement, or AI suggestions with clinical approval.
- Maintain minimum-necessary PHI, human escalation, and safe fallback.